

JEE MAIN REVISION MASTER: COMPOUNDS CONTAINING NITROGEN 2026 3 QUESTIONS 5% WEIGHTAGE

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 27 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Diazonium Salts & Carbylamine Test	18 PYQs	Diazonium & Tests
High	Basicity of Amines & Gabriel Phthalimide Synthesis	9 PYQs	Amine Basics

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Diazonium Salts & Carbylamine Test	$\text{Diazotization: } \text{Ar-NH}_2 + \text{NaNO}_2 + 2\text{HCl} \xrightarrow{0-5^\circ\text{C}} \text{Ar-N}_2^+ \text{Cl}^- + \text{NaCl} + 2\text{H}_2\text{O}$
Basicity of Amines & Gabriel Phthalimide Synthesis	$\text{Basicity in water: } (\text{CH}_3)_2\text{NH} > (\text{CH}_3)_3\text{N} > (\text{CH}_3)_3\text{N}^+ \quad (2^\circ > 1^\circ > 3^\circ)$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Diazonium Salts & Carbylamine Test (Very High Yield — 18 PYQs)

Core Principle: Preparation of benzene diazonium chloride (0-5°C), Sandmeyer/Gattermann halogenation, azo coupling dyes, and Hinsberg separation.

■ **Caution / JEE Pitfall:** Carbylamine test gives foul-smelling isocyanide ONLY with primary amines (1°); secondary and tertiary amines give NO reaction.

■ **Speed Trick:** Hinsberg reagent (C₆H₅SO₂Cl): 1° amine gives product soluble in KOH; 2° amine gives product insoluble in KOH; 3° amine does not react.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "Organic compound 'D' is..." [Ans: N/A]
- [2026 Shift] Q#3 (MCQ): "Nitration of aniline in strong acidic medium gives significant amount of m-nitroaniline because..." [Ans: B]
- [2025 Shift] Q#1 (MCQ): "Which of the following reaction/s does not yield an amine?..." [Ans: N/A]

3.2 Basicity of Amines & Gabriel Phthalimide Synthesis (High Yield — 9 PYQs)

Core Principle: Interplay of inductive, steric, and solvation factors on amine basicity in gas vs aqueous phases, and Hoffmann bromamide degradation.

■ **Caution / JEE Pitfall:** In gas phase, basicity order is purely inductive: $3^\circ > 2^\circ > 1^\circ > \text{NH}_3$. In aqueous solution, solvation and steric factors disrupt this order.

■ **Speed Trick:** Gabriel phthalimide synthesis yields PURE aliphatic primary amines (1°); aromatic amines cannot be prepared because aryl halides do not undergo $\text{S}_{\text{N}}2$.

Representative Source PYQs:

- [2026 Shift] Q#2 (MCQ): "Basic strength of alkylamines in aqueous phase is not decided by _____." [Ans: N/A]
- [2024 Shift] Q#4 (MCQ): "In the reaction, $\text{Aniline} \xrightarrow{\text{Dil.HCl}} \text{NaNO}_2 \xrightarrow{\text{NaOH}} \text{Phenol}$..." [Ans: N/A]
- [2023 Shift] Q#2 (MCQ): "Aniline does not undergo..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

Organic compound 'D' is

- (A)
- (B) CH_3COOH
- (C) CH_3CHO
- (D)

Correct Answer: None

Solution Analysis: Benzene first reacts with concentrated HNO_3 and concentrated H_2SO_4 at 323-333K. This is nitration of benzene, and the product formed is nitrobenzene, marked as (A). $\text{C}_6\text{H}_6 \xrightarrow{\text{Conc. HNO}_3/\text{Conc. H}_2\text{SO}_4} \text{C}_6\text{H}_5\text{NO}_2$ (A) Nitrobenzene (A) is then reduced using Fe/HCl . The nitro group, $-\text{NO}_2$, changes into an amino group, $-\text{NH}_2$. So the product is aniline, marked as (B).
 $\text{C}_6\text{H}_5\text{NO}_2 \xrightarrow{\text{Fe/HCl}} \text{C}_6\text{H}_5\text{NH}_2$

Benchmark Problem 2: KCET 2026 (Q#2)

Basic strength of alkylamines in aqueous phase is not decided by _____.

- (A) Inductive effect
- (B) Solvation effect
- (C) Steric hindrance
- (D) Hyperconjugation effect

Correct Answer: None

Solution Analysis: Basic strength of alkylamines in aqueous phase is not decided by hyperconjugation effect. So, the correct option is: Option D: Hyperconjugation effect. In aqueous solution, the basic strength of alkylamines mainly depends on: Inductive effect: Alkyl groups show +I effect and increase electron density on nitrogen. Solvation effect: The conjugate acid formed after accepting H^+ is stabilised by water molecules. Steric hindrance: Bulky alkyl groups can reduce proper solvation and affect basicity. But hyperconjugation is not an important factor for deciding the basic strength of alkylamine.